NATIONAL ECONOMICS UNIVERSITY

# PROJECT ECONOMETRICS

## RETURNS TO EDUCATION ACROSS SERVICE INDUSTRY SEGMENTS

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**Abstract**

This study analyzes the return on education for wages in the service industry in Vietnam, focusing on clarifying the heterogeneity between two generations (young people: 15-35 years old and middle-aged people: 36-55 years old) and two industry segments (high-end and basic). Based on the 2018 Labor and Employment Survey (LFS) dataset with a sample size of 121,430 observations, the study uses the extended Mincer wage equation combined with OLS estimation with robust standard errors.

Empirical results show that university degrees yield significantly higher returns in the middle-aged group compared to the younger group. Notably, the analysis of interaction variables reveals a clear "saturation effect" among middle-aged workers in high-end service industries, where practical experience and work performance gradually replace the role of formal degrees. Furthermore, the Mincer curve accurately reflects the career lifecycle: a traditional concave shape in the middle-aged group and a linear upward trend in younger workers who are in the experience accumulation phase. These findings provide important implications for policymakers in designing training programs that are aligned with the worker lifecycle and the specifics of each service segment.

# Contents

# 1 Introduction

During Vietnam's ongoing economic restructuring, the service sector has emerged as the primary driver of value creation and employment. However, this sector is far from homogeneous and is currently experiencing profound internal fragmentation. On one end of the spectrum are high-end, knowledge-intensive, and technology-driven services (e.g., finance, information technology). On the other end are low-end, traditional, and labor-intensive services (e.g., retail, accommodation). This structural dichotomy raises a critical question regarding how the labor market prices human capital: Does the wage premium of educational attainment differ significantly between these two distinct service segments?

To address this question, this study utilizes large-scale microdata from the 2018 Vietnam Labor Force Survey (LFS). By extracting key quantitative variables, such as actual monthly income, highest level of educational attainment, and industry classifications. This paper estimates an extended Mincerian wage equation that incorporates an interaction term between education and service segments. Furthermore, the research conducts a targeted subsample analysis on the youth cohort (aged 15–35) and (aged 36 - 55) to evaluate how the digital transformation and the modern economy are reshaping the value of formal educational credentials.

# 2 Rationale

- **Practical and Policy Implications:** The Vietnamese labor market is currently facing the challenge of improving human capital quality to circumvent the middle-income trap. Understanding the wage premium of education within specific service segments provides crucial signals for policymakers and the workforce. If the educational return in low-end services is disproportionately low, it highlights the risk of a skills mismatch and human capital waste, where degree holders are confined to non-specialized jobs. This study offers empirical evidence to better guide career orientation and optimize investments in higher education.
- **Academic Contribution:** While numerous studies have applied the Mincer equation in the context of Vietnam, the majority of previous research estimates the return to education for the aggregate economy or compares the public and private sectors. This study addresses a critical gap in the literature by exploring the internal heterogeneity of the service sector. By categorizing workers into knowledge-intensive versus traditional services, this research provides a more nuanced, targeted, and reliable empirical analysis of wage stratification.

# 3 Theoretical Framework

This study is grounded in three core micro-labor economics theories, which are operationalized to explain wage determination across different service segments:

## 3.1 Human Capital Theory

Pioneered by Gary Becker and Theodore Schultz, this theory posits that formal education and training are investments that enhance a worker's marginal productivity, which the labor market subsequently rewards with higher earnings. In this research, human capital is represented by educational attainment and potential labor market experience, forecasting a positive and statistically significant relationship between education and the natural logarithm of wages.

### 3.2 Signaling or Screening Theory

Proposed by Michael Spence, this theory argues that education does not exclusively directly generate productivity but rather serves as a signal to help employers identify a candidate's innate abilities, thereby reducing information asymmetry. In the high-end service segment, where tasks are complex and immediate productivity is difficult to measure, educational credentials act as a robust signal. Conversely, in low-end services, where productivity is highly observable and tasks are routine, the signaling value of a degree diminishes. This theoretical disparity justifies the introduction of an interaction term between education and service segments into the model.

### 3.3 Labor Market Segmentation Theory

This theory divides the labor market into a primary sector (characterized by stable, high-paying jobs offering upward mobility-analogous to high-end services) and a secondary sector (characterized by precarious, low-paying jobs requiring minimal skills-analogous to low-end services). Wage determination mechanisms differ fundamentally between the two. The secondary sector typically offers marginal wage premiums for additional education. This justifies the comparative approach of the study and the inclusion of variables controlling for employment formality.

### 3.4 Empirical Strategy

Building upon these theoretical foundations, the study employs an extended Mincerian wage equation estimated via Ordinary Least Squares (OLS). To test hypotheses regarding the differential returns to education conditional on the service segment, an interaction term is integrated:

**Sample Regression Function (SRF)**

$$\ln(\text{Wage}_i) = \beta_0 + \beta_1 \textbf{Education}_i + \beta_2 \textbf{Service\_Segment}_i + \beta_3(\textbf{Education}_i \times \textbf{Service\_Segment}_i) + \gamma X_i + \epsilon_i$$

**Where:**

- $\ln(\textbf{Wage}_i)$ is the natural logarithm of the monthly wage.
- $\textbf{Education}_i$ represents educational attainment.
- $\textbf{Service\_Segment}_i$ is a binary variable taking the value of 1 for employment in high-end services and 0 for low-end services.
- $\beta_1$ measures the marginal return to education for workers in the low-end service segment.
- $\beta_2$ measures the intercept shift, representing the baseline wage differential between high-end and low-end service sectors when education is held constant at zero.

- $\beta_3$ is the parameter of primary interest, capturing the differential wage premium of education in high-end services compared to low-end services.
- $X_i$ is a vector of control variables (such as experience, gender, region, and employment contract status) included to mitigate omitted variable bias.
- $\epsilon_i$ is the stochastic error term.

# 4 Rationale for Subsample Analysis: Digital Transformation and Labor Market Dynamics

To capture the changing nature of the labor market, this study divides the dataset into two age groups. This division reflects the service sector's transition from traditional employment to a more digitally integrated economy.

## 4.1 The Young Cohort (Aged 15 - 35) and the Digital Economy

The modern service sector is heavily influenced by digital technology and the rise of the gig economy. Younger workers are much more likely to participate in these new, technology-driven fields, such as tech startups, e-commerce, and digital freelance platforms.

Focusing on this specific group allows us to examine how formal educational credentials are valued in the modern economy. In tech-focused and gig-economy jobs, practical skills and technological adaptability often hold more weight than a traditional degree. Therefore, testing the Mincer equation on this subsample will show whether the financial reward for formal education is changing due to digital transformation. This is particularly relevant when comparing high-end tech jobs to low-end gig work, such as ride-hailing or delivery services.

## 4.2 The Seasoned Cohort (Aged 36 and Above) and Traditional Labor Dynamics

In contrast, older workers generally represent the traditional labor market, which is characterized by standard corporate structures and long-term employment contracts. For this group, the way employers evaluate human capital naturally shifts over time.

According to Human Capital Theory, as workers gain more years of experience, their practical on-the-job skills often become a stronger determinant of their wages than their initial educational attainment. Furthermore, from the perspective of Signaling Theory, the value of a university degree as a "signal" of ability may decline once employers can directly observe a worker's actual productivity over a long period. Analyzing this older cohort allows us to determine if formal degrees still play a significant role in wage differences during the later stages of a career across various service segments.

# 5 Literature Review and Study Rationale

## 5.1 Previous Studies on Returns to Education and Sectoral Heterogeneity

- **Doan (2012) - Return to schooling in Vietnam during economic transition:** This study utilized longitudinal data (1998-2008) to demonstrate that as Vietnam's economy transitioned and the private sector expanded, the rate of return to education increased significantly from under 3% in 1998 to approximately 9-10% in 2008. This indicates the labor market's growing appreciation for human capital.

- **Nguyen (2014) - An analysis of labour market returns to education in Vietnam:** Utilizing the 2012 Labor Force Survey, this comprehensive study identified the "convexity" of the earnings function in Vietnam. Specifically, it found that tertiary education yields a substantially higher wage premium compared to secondary education. However, the study did not deeply explore how this premium varies across specific subsectors within the economy.
- **Groot et al. (2014) - The scope of the external return to higher education:** Analyzing the benefits of higher education in knowledge-intensive sectors, the authors argue that knowledge spillovers and soft skills play a decisive role in high-end services. Consequently, the wage premium for university graduates in these sectors is significantly higher than in traditional sectors.
- **Labor Market Segmentation Literature:** Modern international studies consistently highlight that firm and sector heterogeneity determine the actual value of a degree. In low-end services (e.g., restaurants, retail, delivery), where productivity relies more on physical labor and routine tasks rather than abstract thinking, the signaling value of an academic degree is sharply reduced.

### 5.2 Research Gaps and Study Contributions

Building upon the existing literature, this study is designed to address three primary research gaps in the Vietnamese context:

- **The Lack of an Intra-Industry Perspective:** Most prior research in Vietnam measures the return to education at the aggregate level or compares broad sectors (e.g., agriculture vs. industry vs. services). Few studies recognize the profound internal divide within the service sector itself. This paper fills this gap by introducing an interaction term to capture these intra-industry differences.
- **The Rise of the Gig Economy:** Older studies (pre-2015) do not capture the current reality where a large number of highly educated young workers (with college or university degrees) are entering the low-end service sector (e.g., ride-hailing, express delivery, and small-scale online retail).
- **Underutilization of Detailed Industry Codes:** Many studies overlook the potential of detailed two-digit industry codes in the LFS dataset. This research fully exploits these codes to accurately categorize jobs into high-end and Low-end services, providing a more precise measurement.

### 5.3 Additional Practical Rationales

Beyond the theoretical gaps, this research is driven by several macroeconomic and practical considerations:

- **The 2018 LFS as an Ideal Baseline:** The year 2018 represents a peak in Vietnam's economic growth cycle prior to the COVID-19 pandemic. Using 2018 data provides the cleanest picture of labor market supply and demand, free from the external shocks and massive layoffs of the 2020-2022 period, thereby ensuring high internal validity for our estimates.

- **Addressing the Middle-Income Trap:** Vietnam's growth engine is shifting from cheap, labor-intensive exports to high-value-added services (such as finance, logistics, and ICT). If this study proves that educational returns are vastly superior in high-end services, it delivers a strong policy message: Vietnam must focus on advanced human capital training rather than mass, general education to successfully upgrade its economy.
- **Highlighting the Risk of a Skills Mismatch:** If our empirical results show that the return to education in low-end services is near zero, it highlights a massive social waste. It serves as a data-driven warning about a skills mismatch where students invest years in higher education only to work in roles that do not utilize their specialized knowledge.

# 6 Data and Methodology

## 6.1 Data Preparation and Variable Construction

The data cleaning process focuses on handling missing values, treating outliers, and constructing relevant variables for subsequent analysis.

### 6.1.1 Extracting, Handling Missing Values and Outliers

**Step 1: Data Extraction** (`data_extracting.r`)
Extract raw data from Vietnam LFS 2018 Survey by GSO → `df_raw.csv`

↓

**Step 2: Data Cleaning** (`data_cleaning.r`)
Handle missing values, outliers, and feature engineering → `df_clean.csv`

↓

**Step 3: Data Subsampling** (`data_subsampling.r`)
Split dataset by age groups → `df_young.csv`, `df_middle.csv`

**Preview Dataset:** `df_clean.csv` with sample size n = 121,430 observations

| ln_wage | educ | servseg | educ_servseg | exp | exp_sq | fem | cont | reg_i | age |
|---|---|---|---|---|---|---|---|---|---|
| 6.8565 | 0 | 0 | 0 | 55 | 3025 | 1 | 1 | RRD | 70 |
| 6.9078 | 0 | 0 | 0 | 33 | 1089 | 0 | 1 | RRD | 42 |
| 6.9078 | 0 | 0 | 0 | 31 | 961 | 0 | 1 | RRD | 42 |
| 8.6995 | 0 | 0 | 0 | 12 | 144 | 1 | 1 | RRD | 33 |
| 8.8537 | 0 | 0 | 0 | 37 | 1369 | 1 | 0 | RRD | 55 |

### 6.1.2 Variable Definitions

**Dependent Variable**

- **Logarithm of Wage** (`ln_wage`) - *C44A*: The natural logarithm of monthly income, used to measure percentage changes in earnings and to normalize the right-skewed distribution of wage data.

**Main Independent Variables**

- **Education** (`educ`) - *C17*: A binary variable taking the value of 1 if the worker holds a university degree or higher, and 0 otherwise.
- **Service Segment** (`servseg`) - *C29C*: A binary indicator equal to 1 for employment in high-end, knowledge-intensive services and 0 for low-end, traditional services.

**Interaction Term**

- **Education × Service Segment** (`educ_servseg`) - *C17 × C29C*: Captures the differential wage premium of education across service segments.

**Control Variables ($\mathbf{X}_i$)**

- **Experience** (`exp`) and **Experience Squared** (`exp_sq`) - *C5, C17*: Potential labor market experience (typically calculated as Age − Years of Schooling − 6) and its squared term, included to capture the non-linear (concave) relationship between experience and earnings. Values of `exp` $< 0$ are set to 0.
- **Gender** (`fem`) - *C3*: A binary variable taking the value of 1 for female workers and 0 for male workers, included to control for gender wage differences.
- **Region** (`reg_i`): A set of categorical variables constructed from "TINH", capturing geographical and macroeconomic heterogeneity across Vietnam's six socio-economic regions, with the Red River Delta as the reference group.
- **Contract** (`cont`) - *C36*: A binary variable indicating employment formality (1 = presence of a formal labor contract, 0 = absence of a contract or informal agreement).

## 6.2 Empirical Methodology: Building and Running Model

### 6.2.1 Descriptive Statistics

The summary table below presents the summary statistics for all variables included in the analytical sample, consisting of **121,430 observations**.

The dependent variable, the natural logarithm of monthly wage (*ln_wage*), has a mean value of 8.497, which corresponds to an average monthly income of approximately VND 4,900,000 ($e^{8.497}$), with its near-normal distribution further illustrated in Figure 1. Regarding human capital, the sample indicates that 33.8% of workers hold a university degree or higher (*educ*), while the average labor market experience (*exp*) is 19.9 years. In terms of employment characteristics, 44.6% of the workers are employed in high-end service segments (*servseg*), and 64.2% possess a formal labor contract (*cont*). The interaction term (*educ_servseg*) shows that 30.5% of the sample are highly educated individuals working within high-end services. Finally, the demographic distribution is relatively balanced, with female workers (*fem*) accounting for 48.0% of the total observations.

| Variable | N | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| Log Monthly Wage (*ln_wage*) | 121,430 | 8.497 | 0.567 | 5.704 | 9.976 |
| Education (*educ*) | 121,430 | 0.338 | 0.473 | 0 | 1 |
| Service Segment (*servseg*) | 121,430 | 0.446 | 0.497 | 0 | 1 |
| Educ × Servseg (*educ_servseg*) | 121,430 | 0.305 | 0.461 | 0 | 1 |
| Experience (*exp*) | 121,430 | 19.915 | 12.927 | 0 | 79 |
| Experience Squared (*exp_sq*) | 121,430 | 563.719 | 641.888 | 0 | 6241 |
| Female (*fem*) | 121,430 | 0.480 | 0.500 | 0 | 1 |
| Formal Contract (*cont*) | 121,430 | 0.642 | 0.479 | 0 | 1 |

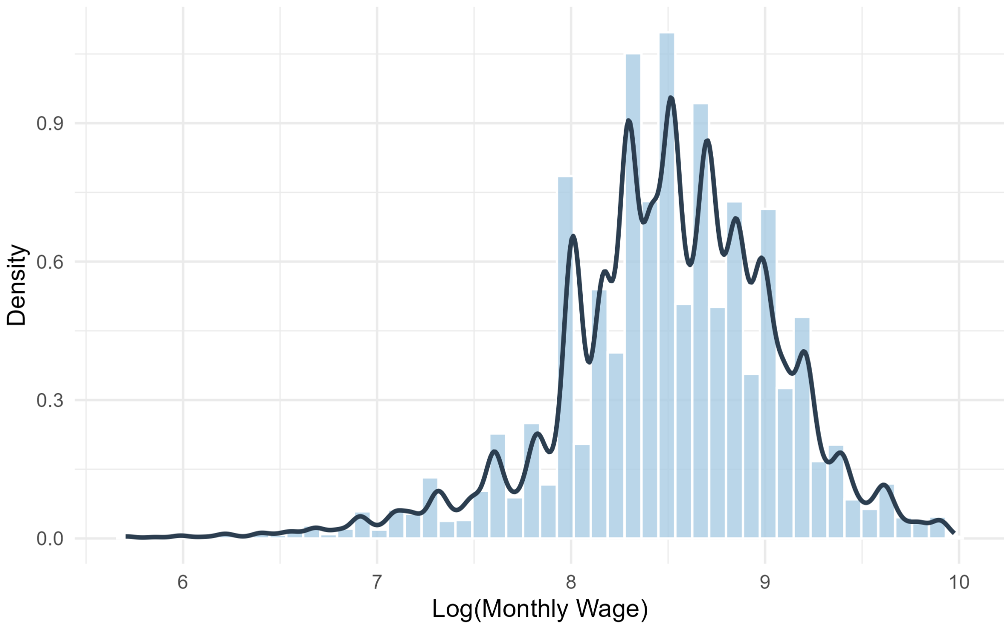


Figure 1: Distribution of Log Monthly Wage

### 6.2.2 Empirical Model

Following the Mincerian approach, the final regression model is specified below:

**Extended Mincerian Wage Equation**

$$\ln(\text{Wage}_i) = \beta_0 + \beta_1\text{educ}_i + \beta_2\text{servseg}_i + \beta_3(\text{educ\_servseg}_i) + \beta_4\text{exp}_i + \beta_5\text{exp}_i^2 + \beta_6\text{fem}_i + \beta_7\text{cont}_i + \sum \delta_j\text{reg\_i}_i + \epsilon_i$$

**Model Overview**

The empirical model estimates the impact of education ***(educ)*** and service sectors (***servseg***) on log wages (***ln_wage***) while controlling for experience (***exp***, ***exp_sq***), gender (***fem***), and regional factors (***reg_i***). The error term ($\epsilon_i$) is included to capture unobserved individual characteristics, such as innate ability or personal motivation, which may influence earnings.

**Subsample Overview**

To capture generational shifts and life-cycle effects, the analysis is further executed across two age-based subsamples: the **Young Cohort (15-35)** and the **Middle-aged Cohort (36-55)**. This stratified approach allows for a more nuanced understanding of how the returns to education and sectoral premiums evolve across different stages of a worker's career.

**Diagnostic Tests**

To ensure the statistical validity and reliability of the OLS estimates, several diagnostic tests are implemented. First, the **Breusch-Pagan test** is conducted to detect **heteroskedasticity**, a common phenomenon in wage data where the variance of the error term is not constant across observations. To address this issue, **Robust Standard Errors** are employed, ensuring that the t-statistics and p-values remain valid for hypothesis testing. Second, the **Variance Inflation Factor (VIF)** is calculated to assess **multicollinearity**. This test ensures that the independent variables are not excessively correlated, which would otherwise lead to unstable and biased coefficient estimates.

# 7 Results and Discussion

## 7.1 Regression Model Results

Table below reports the estimation results of the extended Mincerian wage equation, illustrating how returns to education and sectoral differences vary across age cohorts after controlling for individual and job characteristics.

Table 1: **Estimation Results of Extended Mincerian Wage Equation**

| | **Dependent variable: Log(Wage)** | | |
|---|---|---|---|
| | Full Sample (1) | Young Cohort (15-35) (2) | Middle-aged Cohort (36-55) (3) |
| Education (**educ**) | 0.285*** | 0.233*** | 0.421*** |
| | (0.009) | (0.009) | (0.016) |
| Service Segment (**servseg**) | 0.247*** | 0.099*** | 0.395*** |
| | (0.005) | (0.006) | (0.008) |
| Edu × Service (**educ_servseg**) | -0.072*** | -0.048*** | -0.229*** |
| | (0.010) | (0.010) | (0.016) |
| Experience (**exp**) | 0.029*** | 0.036*** | 0.042*** |
| | (0.0004) | (0.001) | (0.001) |
| Experience Squared (**exp_sq**) | -0.001*** | -0.001*** | -0.001*** |
| | (0.00001) | (0.00004) | (0.00003) |
| Female (**fem**) | -0.140*** | -0.148*** | -0.124*** |
| | (0.003) | (0.004) | (0.004) |
| Formal Contract (**cont**) | 0.235*** | 0.284*** | 0.222*** |
| | (0.004) | (0.005) | (0.006) |
| Constant | 7.921*** | 7.949*** | 7.673*** |
| | (0.005) | (0.007) | (0.021) |
| Region Controls | Yes | Yes | Yes |
| Observations | 121,430 | 58,496 | 54,335 |
| $R^2$ | 0.380 | 0.310 | 0.447 |
| Adjusted $R^2$ | 0.380 | 0.310 | 0.447 |
| Residual Std. Error | 0.446 | 0.413 | 0.422 |
| F Statistic | 6211.758*** | 2186.792*** | 3662.560*** |

*Note: * $p < 0.1$, ** $p < 0.05$,*** $p < 0.01$. Robust standard errors are reported in parentheses.*

### 7.2 Interpretation of Regression Coefficients

Based on the estimation results of the extended Mincerian wage equation, the following sections analyze the impact of socioeconomic factors on wages across the full sample and specific age cohorts. All reported coefficients are statistically significant at the 1 % level ($p < 0.01$) unless otherwise specified.

#### 7.2.1 Education (Degree+)

The coefficient for Education (Degree+) in the Full Sample is 0.285, corresponding to a 33.0% wage increase for degree holders relative to those with lower educational attainment. A cohort-based analysis reveals a stark upward trajectory in the value of education: the premium rises from 26.2% ($\beta = 0.233$) for the Young Cohort to a substantial 52.3% ($\beta = 0.421$) for the Middle-aged. This widening gap in the Middle-aged cohort reflects a divergence in career trajectories, where a degree acts as a prerequisite for accessing senior management roles. It also highlights the income vulnerability of manual laborers once they surpass their physical prime, as their earnings stagnate while the returns for educated professionals continue to compound. This suggests that in the Vietnamese labor market, a university degree acts as more than just a signal for entry-level hiring. Its economic value is compounded over time, likely because degree holders have greater access to internal promotion ladders and senior leadership roles that offer significant salary growth in mid-career stages.

#### 7.2.2 Service Segment (High-end)

Employment in the High-end Service Segment yields a positive coefficient of 0.247 in the Full Sample, representing a 28.0% wage advantage over other sectors. However, the disparity between age groups is substantial: the premium for the Middle-aged Cohort is 48.4% ($\beta$ = 0.395), nearly four times higher than the 10.4% premium ($\beta$ = 0.099) observed for the Young Cohort. This disparity indicates that high-end services (such as finance, technology, and professional consulting) are highly sensitive to human capital accumulation. While the sector provides a baseline benefit for all, it disproportionately rewards industry-specific expertise, professional networks, and soft skills accumulated over time.

#### 7.2.3 Education and Service Interaction (Edu $\times$ Service)

The interaction term explores whether the combination of a degree and employment in high-end services generates a synergistic effect. The coefficients are negative across all models: -0.072 for the Full Sample, -0.048 for the Young Cohort, and -0.229 for the Middle-aged Cohort. This pattern suggests a saturation effect. In high-end sectors, formal education acts as an entry requirement, but its marginal contribution to wages diminishes once individuals are already in these high-paying environments. The combined effect of education and sector is therefore smaller than the sum of their individual effects. For the Middle-aged Cohort, the stronger negative coefficient implies that professional experience and performance may outweigh formal qualifications over time.

#### 7.2.4 Labor Market Experience

The model incorporates both Experience and Experience Squared to capture the wage life-cycle. In the Full Sample, each additional year of experience increases wages by 2.9% ($\beta = 0.029$), while the negative quadratic term (-0.001) indicates diminishing returns. A key distinction emerges across cohorts. The Young Cohort exhibits a near-linear return of 3.7% ($\beta = 0.036$), whereas the Middle-aged Cohort follows a concave earnings profile. This suggests that younger workers are still in a rapid growth phase, while older workers approach their peak earnings.

#### 7.2.5 Gender (Female)

The coefficient for Female is consistently negative across all groups. In the Full Sample, the coefficient of -0.140 implies that women earn 13.1% less than men, holding other factors constant. The gap is larger for the Young Cohort (-13.8%, $\beta = -0.148$) than for the Middle-aged (-11.7%, $\beta = -0.124$). This pattern suggests that structural barriers, such as occupational segregation and glass ceiling effects, remain persistent even among younger workers.

#### 7.2.6 Formal Contract

The presence of a Formal Contract is a strong predictor of higher wages. In the Full Sample, the coefficient of 0.235 corresponds to a 26.5% wage increase. The effect is strongest for the Young Cohort (32.8%, $\beta = 0.284$), compared to 24.9% ($\beta = 0.222$) for the Middle-aged. For younger workers, formal contracts provide access to structured wage systems and social protection, acting as a critical mechanism for upward mobility. For older workers, wage outcomes are more strongly driven by accumulated human capital.

**Summary of Model Fit**

The model demonstrates strong **explanatory power**, particularly for the **Middle-aged Cohort**, with an **Adjusted $R^2$ of 0.447**. This indicates that approximately 45% of wage variation is explained by the included variables.
The **Full Sample and Young Cohort** exhibit lower, but still substantial, explanatory power, with Adjusted $R^2$ values of **0.380 and 0.310**, respectively.

### 7.3 Diagnostic Tests

To ensure the reliability and validity of the estimated coefficients, several diagnostic tests were conducted to address potential econometric issues such as multicollinearity and heteroskedasticity.

#### 7.3.1 Multicollinearity Analysis

The Variance Inflation Factor (VIF) was utilized to detect the presence of multicollinearity among the independent variables. The results indicate relatively high GVIF values for Experience, Experience Squared, Education, and the interaction term (Edu × Service). However, this is identified as structural multicollinearity, which is inherent to the model specification. Specifically, Experience Squared is a deterministic transformation of Experience, and the interaction term is constructed as the product of its underlying variables. As these relationships are intentionally introduced to capture non-linear effects and heterogeneous returns, they do not undermine the validity of the model estimates.

**VIF test result:**

```
GVIF Df GVIF^(1/(2*Df))
educ           8.049971  1         2.837247
servseg        3.066385  1         1.751110
educ_servseg   9.791481  1         3.129134
exp           11.403509  1         3.376908
exp_sq        11.500777  1         3.391280
fem            1.019400  1         1.009653
cont           1.848427  1         1.359569
reg_i          1.086462  5         1.008327
```

#### 7.3.2 Heteroskedasticity Test

To verify the reliability of the OLS estimates, the Breusch-Pagan test is employed to check for the assumption of constant error variance. The formal hypotheses for the test are defined as:

- $H_0 : \sigma_i^2 = \sigma^2$ (Homoskedasticity: The variance of the error terms is constant).
- $H_1 : \sigma_i^2 \neq \sigma^2$ (Heteroskedasticity: The variance of the error terms depends on the independent variables).

The Breusch-Pagan test was conducted on the full sample to evaluate the assumption of homoskedasticity. The test produced a BP statistic of 3568.5 with a p-value $< 2.2 \times 10^{-16}$. Given the near-zero p-value, the null hypothesis of constant variance is strongly rejected, indicating the presence of heteroskedasticity in the error terms. To address this issue, robust standard errors (HC1) were employed for all estimations, ensuring that statistical inference, including t-statistics and significance levels, remains reliable.

**Breusch-Pagan test result:**

```
studentized Breusch-Pagan test

data:  model_full
BP = 3568.5, df = 12, p-value < 2.2e-16
```

### 7.4 Heterogeneity Analysis and Discussion

The impact of human capital and labor market characteristics often varies across different segments of the population. A subsample analysis by age cohorts: Young (15-35) and Middle-aged (36-55), revealing significant heterogeneity in how education and sector choice translate into earnings, as visually summarized in Figure 2.

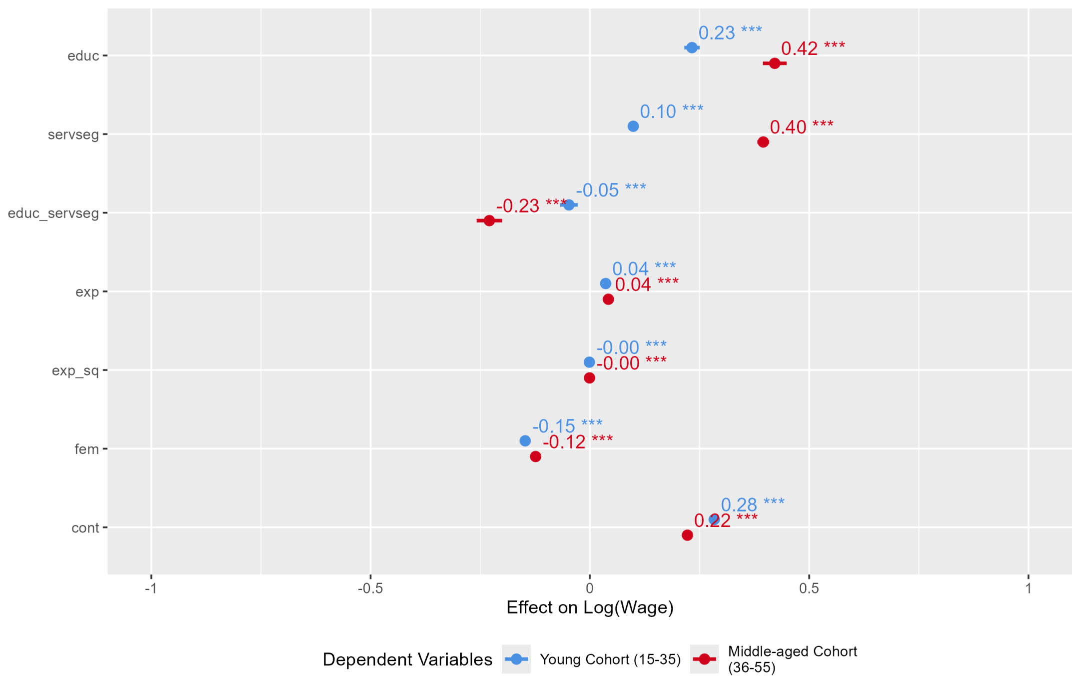


Figure 2: Coefficient Comparison of Wage Determinants between Young and Middle-aged Cohorts

#### 7.4.1 Education Returns Across Cohorts

The returns to education exhibit a marked increase as workers transition from the Young to the Middle-aged cohort. While a university degree increases wages by 23.3% for younger workers, the premium rises substantially to 42.1% for the Middle-aged group ($p < 0.01$). This suggests that a degree functions as more than an entry-level signal; its value is amplified over time, likely because degree holders gain access to career ladders and senior management roles that offer significantly higher compensation as they age.

### 7.4.2 Sectoral Wage Disparities

Employment in the High-end Service Segment provides a wage advantage across both cohorts, but the magnitude differs significantly. For the Young Cohort, working in high-end services increases wages by 9.9%, whereas for the Middle-aged Cohort, the premium rises to 39.5% ($p < 0.01$). This disparity indicates that knowledge-intensive service sectors reward accumulated experience, professional networks, and industry-specific expertise more strongly over time.

### 7.4.3 Education-Sector Interaction

The interaction term Edu $\times$ Service is negative and statistically significant across all models. For the Middle-aged Cohort, the coefficient is $-0.229$, compared to $-0.048$ for the Young Cohort. This suggests a saturation effect: although both education and employment in high-end services increase wages individually, the marginal benefit of a degree is smaller within high-end service sectors than in low-end services. This effect is more pronounced for older workers, where practical experience and performance increasingly dominate formal qualifications, as clearly depicted in Figure 3.

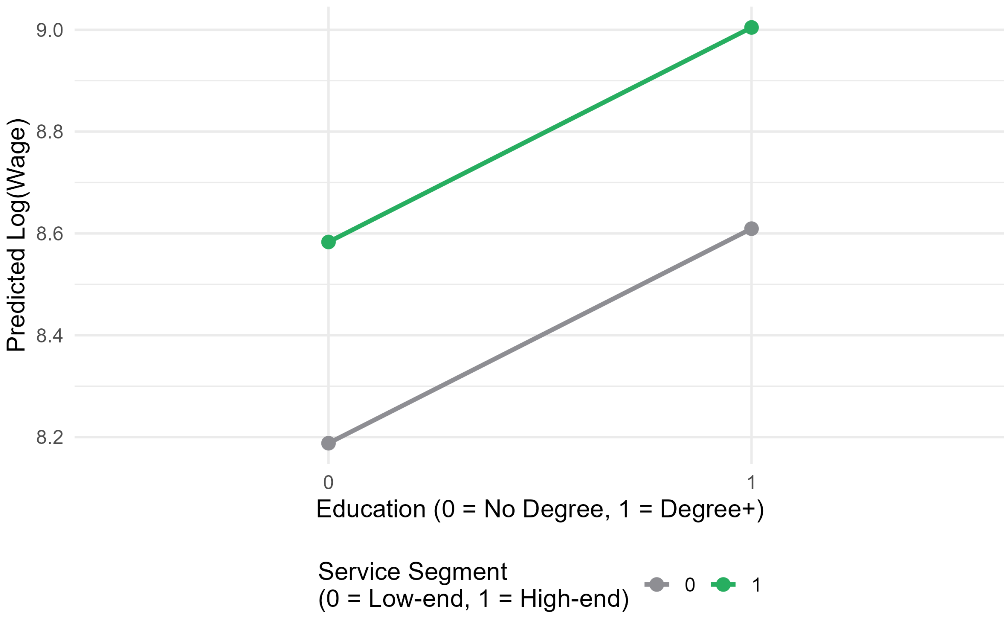


Figure 3: Marginal Effects of Education and Service Segment on Log Wages (Middle-aged Cohort)

### 7.4.4 Career Life-Cycle and Experience

The wage-experience profile confirms a concave relationship for the Middle-aged Cohort, with a positive Experience coefficient (0.042) and a negative Experience Squared term

(−0.001). In contrast, for the Young Cohort, the quadratic term is not statistically significant, suggesting a near-linear wage growth pattern. This indicates that younger workers are still in an early career phase, where returns to experience have not yet begun to diminish, a stark contrast visually captured in the Mincer curves in Figure 4.

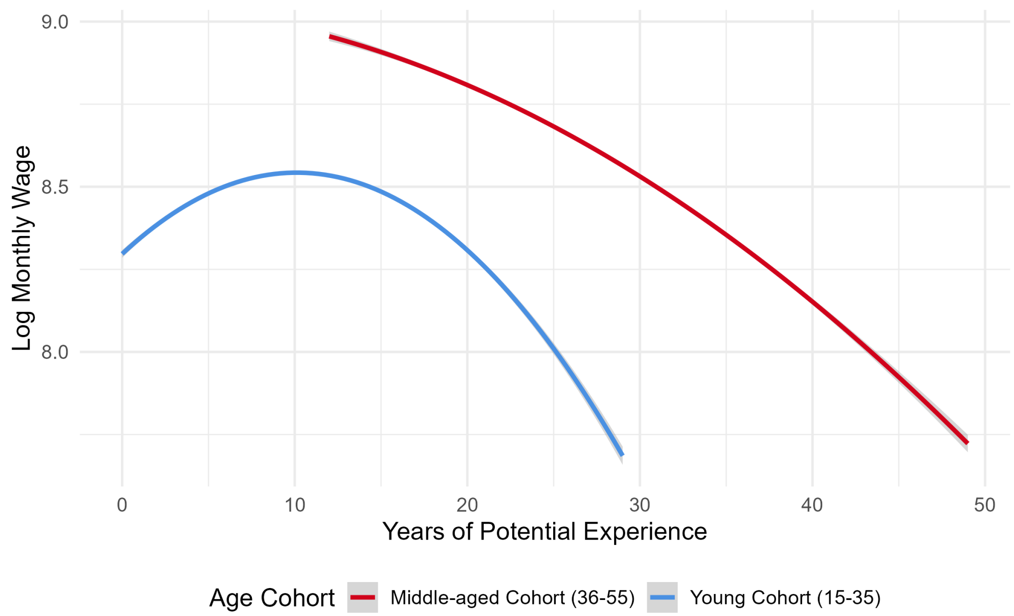


Figure 4: Estimated Wage-Experience Profiles (Mincer Curves) by Age Cohort

#### 7.4.5 Stability and Contract Formality

The presence of a formal labor contract provides a larger wage boost for the Young Cohort (28.4%) than for the Middle-aged Cohort (22.2%) ($p < 0.01$). For younger workers, formal contracts act as a critical mechanism for escaping low-wage informal employment and accessing structured wage systems. While still beneficial for older workers, their earnings are more strongly influenced by accumulated human capital than by contract status alone.

## 8 Conclusion and Policy Implications

### 8.1 Conclusion

**Summary of Findings** This study examined the determinants of wages in Vietnam using an extended Mincerian framework. Based on the empirical analysis of age cohorts and sectoral shifts, three principal findings emerge:

- **Education and Job Formality as Primary Wage Drivers**

  Possession of a university degree and a formal labor contract remain the most robust predictors of high earnings. A formal contract provides a vital wage boost of 26.5% ($\beta = 0.235$) for the general population, while a degree increases wages by 33.0% ($\beta = 0.285$). These factors are especially critical for the younger generation as they seek to escape low-wage, informal employment.

- **Age-Dependent Returns to Education**

  There is a significant experience premium regarding human capital. The financial returns to education nearly double as workers transition from youth to middle age, rising from 26.2% ($\beta = 0.233$) for the Young Cohort to 52.3% ($\beta = 0.421$) for the Middle-aged. This suggests that a degree is not just an entry signal but a foundation for long-term career ladders.

- **Sectoral Disparity and the Seniority Premium**

  Employment in high-end services yields a significant wage advantage, but the benefits are heavily skewed toward older workers. The premium for the Middle-aged Cohort is 48.4% ($\beta = 0.395$), nearly four times higher than the 10.4% premium ($\beta = 0.099$) for the Young Cohort. This indicates that high-end sectors in Vietnam reward industry-specific expertise and professional networks over entry-level participation.

- **The Saturation Effect in Elite Roles**

  The negative interaction between education and high-end services (Edu x Service) across all models indicates a diminishing marginal return. For elite professional roles, especially among middle-aged workers, practical know-how and proven track records appear to become more impactful than formal schooling alone, leading to a slight saturation of the degree's incremental value in these sectors.

- **Non-Linear Career Life-Cycle**

  The study confirms a concave relationship between experience and earnings. While younger workers enjoy a linear wage growth of 3.7% per year, the growth rate for middle-aged workers slows as they approach their peak earning potential. This highlights a critical phase for mid-career workers to update skills before reaching the "plateau" of their professional life-cycle.

- **Persistent Gender Wage Gap**

  Despite modernization, a significant gender wage gap persists. Women earn 13.1% less than men holding similar qualifications ($\beta = -0.140$). Notably, this gap is wider among the Young Cohort (-13.8%) than the Middle-aged (-11.7%), suggesting that increased educational attainment among younger women has not yet managed to dismantle deep-seated structural barriers and glass ceilings.

### 8.2 Policy Implications

Based on the findings of this study, the following policy recommendations are proposed to enhance wage levels and labor market equity in Vietnam, aligned with the nation's current legal framework.

1. **Modernizing Higher Education and Industry Alignment**

   To close the gap between academic training and market demand, the government should incentivize university-industry partnerships. Aligned with the 2018 Law on Higher Education, this practical integration ensures young graduates acquire specific technical skills to secure higher early-career wage premiums.

2. **Strengthening Labor Formality and Administrative Reform**

   Formal contracts act as critical wage elevators. In line with the 2019 Labor Code and Draft Law on Social Insurance (2024), policymakers should simplify administrative procedures and reduce compliance costs for SMEs. This will encourage businesses to formalize employment, extending structured pay scales to more workers.

3. **Closing the Gender Wage Gap and Supporting Work-Life Balance**

   Despite existing legal frameworks, structural wage disparities persist. Following Decree 145/2020/NĐ-CP and the National Strategy on Gender Equality, the government must expand affordable childcare. Relieving women of unpaid care burdens enables them to compete fairly for high-paying, knowledge-intensive roles.

4. **Institutionalizing Lifelong Learning for a Digital Economy**

   To prevent wage stagnation among middle-aged workers during Industry 4.0, the state should finance digital upskilling programs using the Unemployment Insurance Fund (Law on Employment 2013). This allows the aging workforce to leverage their extensive experience alongside new technological skills.

### 8.3 Limitations and Future Research

While this study provides valuable insights, it has certain limitations. The research uses data from a specific point in time, which makes it difficult to track how individual wages change over many years. Future studies should try to follow the same group of people over a longer period to better understand career progression. Additionally, the category of high-end services used here is quite broad. Future research could break this down into more specific fields, like banking or software engineering, to see exactly which industries offer the best returns. Finally, as remote work and digital platforms become more common in Vietnam, future models should include variables related to digital literacy and technology use to see how these new factors influence modern wages.

## References


[1] Doan, T., & Gibson, J. (2012). *Return to schooling in Vietnam during economic transition*. Economics of Education Review.

[2] Doan, T., Ha, V., Tran, T., & Yang, J. (2023). *Dynamics of wage inequality over the prolonged economic transformation: The case of Vietnam*. Economic Analysis and Policy, 78, 816–834. https://doi.org/10.1016/j.eap.2023.04.013

[3] Government of Vietnam. (2020). *Decree No. 145/2020/ND-CP on labor conditions and labor relations*. Government Office.

[4] Groot, W., & Maassen van den Brink, H. (2014). *The scope of the external return to higher education*. European Journal of Education.

[5] National Assembly of Vietnam. (2006). *Law on Gender Equality (Law No. 73/2006/QH11)*. National Assembly House.

[6] National Assembly of Vietnam. (2013). *Law on Employment (Law No. 38/2013/QH13)*. National Assembly House.

[7] National Assembly of Vietnam. (2018). *Law on Higher Education (Law No. 34/2018/QH14)*. National Assembly House.

[8] National Assembly of Vietnam. (2019). *Labor Code (Law No. 45/2019/QH14)*. National Assembly House.

[9] Nguyen, H. C. (2014). *An analysis of labour market returns to education in Vietnam*. Development Policy Review.

[10] Nguyen, C. V. (2021). *Returns to education in Vietnam: A changing landscape*. World Development, 138, 105205. https://doi.org/10.1016/j.worlddev.2020.105205

[11] Nguyen, C. V., Chacaltana, J., Estupinan, X., & Huynh, P. (2025). *Heterogeneous returns to education in labor markets: New evidence from Vietnam*. International Journal of Educational Development, 118, 103407. https://doi.org/10.1016/j.ijedudev.2025.103407

[12] Prime Minister of Vietnam. (2021). *Decision No. 1373/QD-TTg on Building a learning society 2021-2030*. Government Office.

[13] World Bank. (2024). *Vietnam: Investing in Skills to Accelerate Job Transitions*. World Bank Publications. https://openknowledge.worldbank.org/handle/10986/41232

[14] Becker, G. S. (1964). *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*. University of Chicago Press.

[15] Doeringer, P. B., & Piore, M. J. (1971). *Internal Labor Markets and Manpower Analysis*. Heath Lexington Books.

[16] General Statistics Office (GSO). (2019). *Report on the Vietnam Labor Force Survey 2018*. Statistical Publishing House.

[17] Lemieux, T. (2006). The "Mincer equation" thirty years later. In *Jacob Mincer: A Pioneer of Modern Labor Economics* (pp. 127–148). Springer.

[18] Mincer, J. (1974). *Schooling, Experience, and Earnings*. National Bureau of Economic Research (NBER).

[19] Nguyen, V. C. (2013). Return to Education in Vietnam: Evidence from Household Surveys. *Vietnam's Socio-Economic Development*.

[20] Sakellariou, C., & Fang, Z. (2014). Returns to Schooling in Vietnam during Economic Transition: 1994–2009. *Post-Communist Economies, 26*(1), 54–69.

[21] Spence, M. (1973). Job Market Signaling. *The Quarterly Journal of Economics, 87*(3), 355–374.

[22] World Bank. (2021). *Vietnam – Vibrant Vietnam: Forging the Strategies of Advanced Manufacturing and High-End Services*. World Bank Group.

# A Appendix: R Source Code

## A.1 Raw Data Extraction

```
# Load required libraries
library(haven)
library(tidyverse)
library(readr)

# Import raw data from Stata file (.dta)
df_raw <- read_dta("data/dta/LFS_2018 (3)_cut_dup.dta")

# Select relevant variables from the dataset
df_selected <- df_raw %>%
    select(TINH, C3, C5, C17, C29C, C36, C44A)

# Preview the first rows of the selected data
head(df_selected)

# Export the selected data to CSV format
write_csv(df_selected, "data/csv/df_raw.csv")

# Free memory by removing the raw dataset
rm(df_raw)
gc()    # Run garbage collection to release memory
```

## A.2 Data Cleaning and Variable Construction

```
# Load required libraries
library(tidyverse)
library(stringr)
library(readr)

# Import raw dataset from CSV
df <- read_csv("data/df_raw.csv")

# Calculate 1% and 99% quantile bounds for wage (exclude non-
   positive values)
lower_bound <- quantile(df$C44A[df$C44A > 0], 0.01, na.rm = TRUE)
upper_bound <- quantile(df$C44A[df$C44A > 0], 0.99, na.rm = TRUE)

# Data cleaning pipeline and variable creation
df_clean <- df %>%

  # STEP 1: Filter missing values, apply wage bounds, and keep
     labor age >= 15
  filter(
    !is.na(C44A), C44A >= lower_bound, C44A <= upper_bound,
    !is.na(C17),
    !is.na(C29C),
    !is.na(C5), C5 >= 15,
    !is.na(C36),
```

```
    !is.na(C3),
    !is.na(TINH)
  ) %>%

  # STEP 2: Create new variables
  mutate(
    # Dependent variable: log of wage
    ln_wage = log(C44A),

    # Education dummy: 1 = university, 0 = others
    educ = ifelse(C17 %in% c(8, 9), 1, 0),

    # Service segment classification from occupation code
    level1 = substr(as.character(C29C), 1, 1),
    servseg = case_when(
      level1 %in% c("1", "2", "3") ~ 1, # High-end
      level1 %in% c("4", "5", "9") ~ 0, # Low-end
      TRUE ~ NA_real_
    ),

    # Interaction term: education    service segment
    educ_servseg = educ * servseg,

    # Years of schooling based on education code
    years_of_schooling = case_when(
      C17 == 1 ~ 0, C17 == 2 ~ 3, C17 == 3 ~ 5, C17 == 4 ~ 9,
      C17 == 5 ~ 12, C17 == 6 ~ 14, C17 == 7 ~ 15, C17 == 8 ~ 16,
         C17 == 9 ~ 18,
      TRUE ~ 0
    ),

    # Experience = Age - schooling years - 6, capped at 0
    exp = C5 - years_of_schooling - 6,
    exp = ifelse(exp < 0, 0, exp),
    exp_sq = exp^2,

    # Gender dummy: 1 = female, 0 = male
    fem = ifelse(C3 == 2, 1, 0),

    # Contract dummy: 1 = formal contract, 0 = informal/no contract
    cont = ifelse(C36 %in% c(1, 2, 3, 4, 5), 1, 0),

    # Region grouping into 6 socio-economic regions
    region_group = case_when(
      TINH %in% c(1, 17, 26, 27, 30, 31, 33, 34, 35, 36, 37) ~ "RRD
         ",
      TINH %in% c(2, 4, 6, 8, 10, 11, 12, 14, 15, 19, 20, 22, 24,
         25) ~ "NMM",
      TINH %in% c(38, 40, 42, 44, 45, 46, 48, 49, 51, 52, 54, 56,
         58, 60) ~ "Coast",
      TINH %in% c(62, 64, 66, 67, 68) ~ "Highlands",
      TINH %in% c(70, 72, 74, 75, 77, 79) ~ "SE",
      TINH %in% c(80, 82, 83, 84, 86, 87, 89, 91, 92, 93, 94, 95,
         96) ~ "MRD",
      TRUE ~ NA_character_
    ),
```

```
    reg_i = factor(region_group),
    reg_i = relevel(reg_i, ref = "RRD") # Red River Delta as
        baseline
  ) %>%

  # STEP 3: Keep only valid service segment observations
  filter(!is.na(servseg)) %>%

  # Select variables for regression analysis
  select(ln_wage, educ, servseg, educ_servseg, exp, exp_sq, fem,
      cont, reg_i, age = C5)

# Save cleaned dataset to CSV
write_csv(df_clean, "data/csv/df_clean.csv")

# Summary statistics of cleaned data
summary(df_clean)
```

### A.3 Age Cohort Subsampling

```
# Load required libraries
library(tidyverse)
library(readr)

# Import cleaned dataset
df_clean <- read_csv("data/csv/df_clean.csv")

# Subsample 1: Young cohort (ages 15  35  )
df_young <- df_clean %>%
    filter(age >= 15 & age <= 35)

# Save young cohort data and check sample size
write_csv(df_young, "data/csv/df_young.csv")
nrow(df_young)

# Subsample 2: Middle-aged cohort (ages 36  55  )
df_middle <- df_clean %>%
    filter(age > 35 & age <= 55)

# Save middle-aged cohort data and check sample size
write_csv(df_middle, "data/csv/df_middle.csv")
nrow(df_middle)
```

### A.4 Empirical Modeling and Diagnostic Tests

```
# LOAD NECESSARY LIBRARIES AND DATASETS

# Required libraries for regression and diagnostics
library(readr)
library(tidyverse)
```

```
library(car)        # Variance Inflation Factor (VIF) test
library(lmtest)     # Breusch-Pagan test for heteroskedasticity
library(sandwich)   # Robust standard errors
library(stargazer)  # Regression output tables

# Load datasets (full sample, young cohort, middle-aged cohort)
df_full <- read_csv("data/df_clean.csv")
df_young <- read_csv("data/df_young.csv")
df_middle <- read_csv("data/df_middle.csv")

# Convert region variable to factor type
df_full$reg_i <- as.factor(df_full$reg_i)
df_young$reg_i <- as.factor(df_young$reg_i)
df_middle$reg_i <- as.factor(df_middle$reg_i)

# REGRESSION MODELS
# Define regression formula (extended Mincerian wage equation)
formula_model <- ln_wage ~ educ + servseg + educ_servseg + exp +
    exp_sq + fem + cont + reg_i

# Estimate models for full sample, young cohort, and middle-aged
    cohort
model_full <- lm(formula_model, data = df_full)
model_young <- lm(formula_model, data = df_young)
model_middle <- lm(formula_model, data = df_middle)

# DIAGNOSTIC TESTS

# 1. VIF Test (check multicollinearity)
print("VIF Test for Full Sample:")
vif_results <- vif(model_full)
print(vif_results)

# 2. Breusch-Pagan Test (check heteroskedasticity)
print("Breusch-Pagan Test for Full Sample:")
# H0: Homoscedasticity. Reject H0 if p-value < 0.05.
bp_test <- bptest(model_full)
print(bp_test)

# EXPORT RESULTS WITH ROBUST STANDARD ERRORS

# Compute robust standard errors for each model
robust_se_full <- sqrt(diag(vcovHC(model_full, type = "HC1")))
robust_se_young <- sqrt(diag(vcovHC(model_young, type = "HC1")))
robust_se_middle <- sqrt(diag(vcovHC(model_middle, type = "HC1")))

# Generate regression tables with stargazer (HTML output)
stargazer(model_full, model_young, model_middle,
          type = "html",
          out = "docs/Regression_Results.doc",
          title = "Table 1: Estimation Results of Extended
             Minceriian Wage Equation",
          column.labels = c("Full Sample", "Young Cohort (15-35)",
             "Middle-aged Cohort (36-55)"),

          # Insert robust standard errors
```

```
          se = list(robust_se_full, robust_se_young, robust_se_
             middle),

          dep.var.labels = "Log(Wage)",
          covariate.labels = c("Education (Degree+)", "Service
             Segment (High-end)",
                               "Edu x Service", "Experience", "
                                  Experience Squared",
                               "Female", "Formal Contract"),

          omit = "reg_i",
          add.lines = list(c("Region Controls", "Yes", "Yes", "Yes"
             )),

          star.cutoffs = c(0.1, 0.05, 0.01),
          notes = "Note: Robust standard errors are reported in
             parentheses.",
          notes.align = "l")
```

## A.5 Data Visualization and Chart Generation

```
# Load required libraries
library(tidyverse)
library(ggplot2)
library(sjPlot)     # For regression plots
library(ggeffects)  # For marginal effects

# Create folder for plots if not exists
if(!dir.exists("plots")) dir.create("plots")

# [1] Wage distribution plot
p1 <- ggplot(df_full, aes(x = ln_wage)) +
  geom_histogram(aes(y = ..density..), bins = 50, fill = "#A9CCE3",
      color = "white", alpha = 0.8) +
  geom_density(color = "#2C3E50", size = 1.2) +
  labs(x = "Log(Monthly Wage)", y = "Density") +
  theme_minimal(base_size = 14)

# Save histogram
ggsave("plots/01_wage_distrubution.png", plot = p1, width = 8,
   height = 5, dpi = 300)

# [2] Forest plot comparing young vs. middle-aged cohorts
col_middle = "#D0021B"
col_young = "#4A90E2"
p2 <- plot_models(model_young, model_middle,
                  m.labels = c("Young Cohort (15-35)", "Middle-aged
                     Cohort (36-55)"),
                  show.values = TRUE, show.p = TRUE,
                  dot.size = 2.5, line.size = 1,
                  rm.terms = c("reg_iHighlands", "reg_iMRD", "reg_
                    iNMM", "reg_iRRD", "reg_iSE", "RRD", "SE"),
                  title = "",
```

```
                    axis.title = "Effect on Log(Wage)",
                    colors = c(col_middle, col_young)) +
  theme(legend.position = "bottom")

# Save forest plot
ggsave("plots/02_forest_plot.png", plot = p2, width = 9, height =
    6, dpi = 300)

# [3] Marginal effects of education    service segment
interact_pred <- ggpredict(model_middle, terms = c("educ", "servseg
    "))
df_interact <- as.data.frame(interact_pred)

p3 <- ggplot(df_interact, aes(x = as.factor(x), y = predicted,
    color = group, group = group)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  labs(x = "Education (0 = No Degree, 1 = Degree+)",
       y = "Predicted Log(Wage)",
       color = "Service Segment \n(0 = Low-end, 1 = High-end)") +
  scale_color_manual(values = c("0" = "#8E8E93", "1" = "#27AE60"))
    +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")

# Save interaction effect plot
ggsave("plots/03_interaction_effect.png", plot = p3, width = 8,
    height = 5, dpi = 300)

# [4] Mincer curve (wage-experience profile)
p4 <- ggplot() +
    geom_smooth(data = df_young, aes(x = exp, y = ln_wage, color =
       "Young Cohort (15-35)"),
                method = "lm", formula = y ~ poly(x, 2), se = TRUE,
                    size = 1.2) +
    geom_smooth(data = df_middle, aes(x = exp, y = ln_wage, color =
        "Middle-aged Cohort (36-55)"),
                method = "lm", formula = y ~ poly(x, 2), se = TRUE,
                    size = 1.2) +
    labs(x = "Years of Potential Experience", y = "Log Monthly Wage
       ", color = "Age Cohort") +
    scale_color_manual(values = c("Young Cohort (15-35)" = "#4A90E2
       ",
                                  "Middle-aged Cohort (36-55)" = "#
                                     D0021B")) +
    theme_minimal(base_size = 14) +
    theme(legend.position = "bottom")

# Save Mincer curve plot
ggsave("plots/04_mincer_curve.png", plot = p4, width = 8, height =
    5, dpi = 300)
```